NAME: _____ Geometry (FL B.E.S.T.)

## 9.2 Inscribed Angles

An **inscribed angle** has its vertex on the circle and its two sides are chords. The **Inscribed Angle Theorem** says an inscribed angle is always *half* its intercepted arc: $m\angle = \frac{1}{2} m\overset{\frown}{\text{arc}}$. Two immediate consequences: inscribed angles that intercept the *same* arc are congruent to each other, and an inscribed angle is half of the central angle on the same arc (since the central angle equals the arc). A special case called **Thales' Theorem** follows directly: if a triangle is inscribed in a circle with one side as a diameter, the angle opposite that diameter is always a right angle, because a diameter's arc is $180°$ and half of $180°$ is $90°$. This worksheet includes the reasoning behind the theorem, not just its use.

## Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

**1.** An inscribed angle intercepts an arc of $110°$. Find the angle.

**2.** An inscribed angle measures $37°$. Find its intercepted arc.

**3.** A central angle measures $150°$. Find the inscribed angle on the same arc.

**4.** Two inscribed angles on the same arc measure $(4x - 6)°$ and $(2x + 10)°$. Find $x$ and the angle.

**5.** Diameter $\overline{PQ}$ with $R$ on the circle. $m\angle P = 42°$. Find $m\angle Q$.

**6.** Inscribed angle intercepts a $68°$ arc. Find the angle.

**7.** Inscribed angle is $81°$. Find the intercepted arc.

**8.** Central angle is $88°$. Find the inscribed angle on the same arc.

**9.** Inscribed triangle with diameter hypotenuse; acute angles $3x$ and $2x$. Find $x$ and both angles.

**10.** Angles $(x + 12)°$ and $(3x - 8)°$ intercept the same arc. Find $x$ and the angle.

**11.** Arc measure $140°$. Find the inscribed angle intercepting it.

**12.** Inscribed angle $49°$. Find the arc.

**13.** Where is the vertex of an inscribed angle located?

**14.** Two inscribed angles intercept arcs that together make up the whole circle, and one angle is 3 times the other. Find both inscribed angles.

**15.** A right triangle is inscribed with the hypotenuse as a diameter of length 26. One leg is 24. Find the other leg and both acute angles (nearest degree).

## ✓ Answer Key

1. 55°
2. 74°
3. 75°
4. $x = 8$; angle 26°
5. 48°
6. 34°
7. 162°
8. 44°
9. $x = 18°$; angles 54°, 36°
10. $x = 10$; angle 22°
11. 70°
12. 98°
13. On the circle
14. 135° and 45°
15. Other leg = 10; angles ≈ 67° and ≈ 23°